Oscillometric Blood Pressure Estimation via Machine Learning Analysis of Pulse Transit Time

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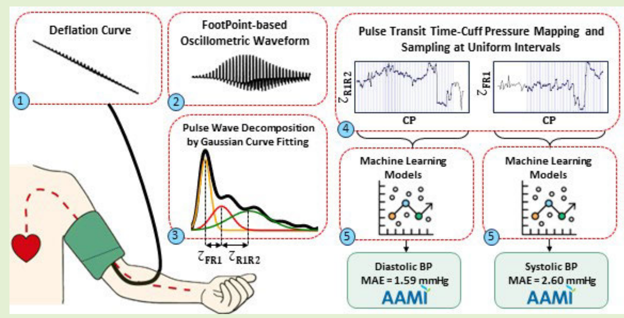
***Abstract*—Oscillometry, a common noninvasive technique, traditionally estimates blood pressure (BP) from the pattern of arterial pulse amplitudes at different cuff pressures (CPs). However, accuracy can be limited with irregular pulses, such as in obese, arteriosclerotic, and atrial fibrillation patients. To address this, this article introduces a novel BP estimation approach that focuses on the timing of oscillometric pulses. We use Gaussian curve fitting (GCF) to decompose each pulse waveform into its forward and reflected components. This decomposition allows us to accurately extract pulse transit time (PTT) from the intervals between waveform components. Leveraging machine learning (ML), we model the relationship between systolic and diastolic BPs and PTT patterns relative to CP. We evaluated this method on two datasets: 425 wrist and 150 upper arm recordings. The results show improved accuracy compared to previous work, with the best results achieved using a random forest model, yielding mean absolute errors (MAEs) of 2.60 and 1.59 mmHg for systolic and diastolic BP in the first dataset and 4.97 and 2.80 mmHg in the second dataset. Our approach met the Association for the Advancement of Medical Instrumentation (AAMI) and British Hypertension Society (BHS) standards for both systolic and diastolic BP, offering enhanced accuracy and reliability for oscillometric BP devices and potential applicability in challenging patient populations.**



***Index Terms*— Blood pressure (BP) estimation, machine learning (ML), oscillometry, pulse decomposition, pulse transit time (PTT).**

I. Introduction

Accurate blood pressure (BP) measurement is vital in medicine. While highly accurate, invasive intra-arterial BP monitoring carries risks. Therefore, noninvasive methods like palpation, auscultation, and oscillometry are preferred for their safety and ease of use [1], [2], [3]. Auscultation, using a sphygmomanometer and stethoscope, is a reliable standard requiring no regular calibration. However, oscillometry has become increasingly popular in automatic BP monitors due to its ease of automation for measuring systolic (SBP), diastolic (DBP), and mean arterial pressure (MAP), differing from auscultation by utilizing a pressure sensor under the cuff to accurately record pressure changes during deflation [3]. The resultant "cuff deflation curve" captures the relationship between the applied cuff pressure (CP) and the arterial oscillations (pulses) that can be extracted and analyzed. These arterial oscillations, forming the "oscillometric waveform (OMW)," vary in amplitude at different CPs and correlate with BP levels [2].

Various methods have been developed to estimate BP using oscillometric pulse amplitude patterns. These often involve empirical coefficients for estimating BP from the OMW envelope (OMWE) [4], [5] or the analysis of time-domain features of the waveform, such as pulsewidth and slope [6], [7]. Recently, researchers have explored a variety of machine learning (ML) approaches for oscillometric BP estimation. One approach utilizes feature-based neural networks to analyze specific characteristics within OMW [8]. Another technique combines 2-D oscillometric data representation with

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hybrid convolutional-recurrent neural networks to capture both waveform shape and temporal changes, enabling accurate BP estimation [9]. Additionally, studies have employed deep belief network–deep neural networks (DBN-DNNs) in conjunction with fusion ensemble regression models, utilizing techniques like bootstrap aggregation and AdaBoost for enhanced performance [10], [11]. Furthermore, a novel regression technique using deep Boltzmann regression with mimic features has been developed to effectively capture complex, nonlinear relationships between oscillometric signals and target BPs [12]. Long short-term memory-recurrent neural networks (LSTM-RNNs) have also been employed to analyze time-dependent patterns extracted from beat-by-beat features within OMWs [13]. Building upon this, a subsequent study [14] introduced a DBN-DNN classification model with a novel bootstrapping method to generate artificial features, further improving classification accuracy.

However, these amplitude-based methods encounter challenges, particularly when dealing with conditions like arterial fibrillation, obesity, and atherosclerosis, where pulse amplitudes can be weak or irregular [15], [16]. Muscle movements and other artifacts can also disrupt accurate pulse amplitude recording, as noted in [17].

Recognizing these challenges, recent researchers have explored the relationship between pulse transit time (PTT) and both cuffless [18], [19], [20] and cuff-based [21], [22] BP estimation methods. In oscillometry, PTT is measured as the interval between characteristic points on oscillometric pulses and ECG R-peaks, with distinct PTT-CP patterns enabling estimation of SBP and DBP [22]. However, reliance on multiple sensors (e.g., ECG and pressure sensors) presents challenges for wearable devices, as synchronizing and processing these signals in real time adds computational demand and affects battery efficiency. Additionally, minimizing the number of sensors is essential for reducing costs and improving battery life, which supports the practicality of wearable BP monitors.

We introduce a fundamentally new approach for oscillometric BP estimation using only the cuff deflation curve. We focus on the analysis of specific PTT patterns, shifting focus away from the conventional approach based on oscillometric pulse amplitude patterns. Our method analyzes the behavior of arterial pulses within the cardiovascular system. Every pulse interacts with changes in arterial resistance, generating a combined wave that includes both transmitted and reflected components. We propose dissecting these waves into their constituent subwaveforms, offering a new perspective on how to interpret PTT. Specifically, we calculate PTTs as the intervals between peaks of these dissected subwaveforms, investigating their correlation with CP. The concept of using PTT between oscillometric subwaveforms for BP estimation was initially investigated in [21], which identified a relationship between specific PTT patterns and BP. However, existing approaches often rely heavily on derivative calculations, which are sensitive to noise, leading to significant inaccuracies. Furthermore, the inherently noisy and inconsistent nature of PTT patterns presents challenges in establishing a direct and reliable relationship with BP. To address these challenges, this study introduces key innovations to enhance BP estimation.

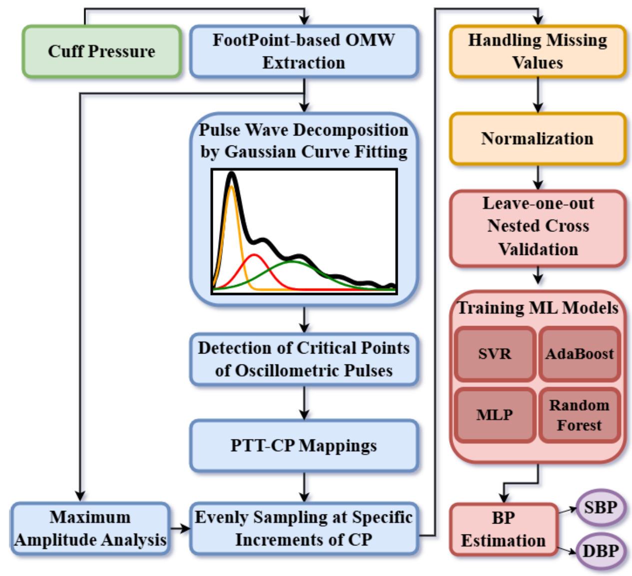

Fig. 1. Workflow of the proposed oscillometric BP estimation method using pulse decomposition and ML.

1) *GCF for Oscillometric Pulse Decomposition:* We present a novel technique for decomposing oscillometric pulses into forward and reflected waves using Gaussian curve fitting (GCF). This approach provides a flexible framework that accommodates waveform variability while minimizing least-squares error (LSE), reducing sensitivity to noise. Decoupling reliance on precise derivative values enhances the stability of PTT estimates.
2) *ML Integration:* To model the complex relationship between noisy PTT patterns and BP, we propose a robust ML framework. By systematically evaluating a range of algorithms, including random forest, AdaBoost, support vector regression (SVR), and multilayer perceptron (MLP), we aim to identify the most accurate and reliable model for BP estimation.

We rigorously evaluate our proposed methods on two distinct oscillometric datasets and verify their performance against the Association for the Advancement of Medical Instrumentation (AAMI) [23] and the British Hypertension Society (BHS) [24] standards for both SBP and DBP estimation.

II. Methodology

Our methodology begins by extracting the OMW from the cuff deflation curve. We then decompose each oscillometric pulse into its constituent components—the forward waveform (FW) and two reflected waveforms (RWs)—using GCF. Next, we calculate the PTT as the interval between critical points of the FW and RWs, establishing its relationship to CP. Finally, we employ ML algorithms to model the relationship between these PTT patterns and BP. For a visual representation of this workflow and its key steps, please refer to Fig. 1.

A. Dataset

To assess the performance of our proposed method for BP estimation, we used two distinct oscillometric datasets recorded over the arm and wrist.

The first dataset consisted of 150 simultaneous recordings of BP and ECG using a prototype designed at the University of Ottawa [25]. Data were recorded from ten healthy individuals, including four women aged 24–63 years, at a sampling frequency of 256 Hz. This research obtained approval from the University of Ottawa Research Ethics Board (Reference Number H02-10-01). Written consent was obtained from all participants; none reported a history of cardiovascular or respiratory disease. Recordings were made on three separate days, with five sessions per day. BP measurements were obtained in the following order: first, using the FDA-approved Omron device (HEM-790IT) on the right arm, followed by the prototype measurement on the left arm. Simultaneously, a wristband worn on the right wrist recorded the ECG. In this study, we focused exclusively on the cuff deflation curve. The measured SBP and DBP ranged from 79 to 136 mmHg and from 52 to 86 mmHg, respectively. More details about this dataset can be found in [25].

The second dataset, provided by Biosign Technologies Inc., was recorded using a wrist BP monitor (UFIT TEN-10) at a sampling frequency of 100 Hz. The research obtained ethical approval from a local Research Ethics Board (Reference Number B-2008-08-17) and adhered to the guidelines outlined by the American National Standards Institute (ANSI)/AAMI SP10: 2002 standard [26]. It includes data from 85 subjects, including 37 women aged 12–80 years. Five sets of oscillometric BP measurements were obtained from each subject (425 sets total); 1 min after each wrist measurement, two nurses measured SBP and DBP using the auscultatory method. The average of these two measurements served as the reference BP for comparison. The recorded SBP ranged from 78 to 147 mmHg, and DBP ranged from 42 to 99 mmHg. Further information is available in [8].

B. OMW Extraction

To achieve a precise estimate of the OMW unaffected by filtering artifacts and suitable for accurate PTT estimation, we implemented a systematic extraction process based on the foot point-based method introduced in our previous study [21]. This method involves the following key steps.

1) Initial Estimation: The OMW was initially estimated by removing low-frequency trends from the cuff deflation curve, which could obscure oscillatory features, using a 0.5-Hz high-pass Butterworth filter. This cutoff frequency was selected based on typical oscillometric signal characteristics [2]. The resulting waveform, referred to as the “filtered-based OMW,” serves as the initial representation of the underlying oscillatory components.

2) Peak Identification: The peaks of the filtered-based OMW were identified using the automatic multiscale-based peak detection (AMPD) technique [27]. This method eliminates the need for predefined thresholds or window lengths, enabling accurate detection of peaks in both periodic and quasi-periodic signals.

3) Foot Point Determination: For each pair of consecutive peaks, the “foot points” were identified as the locations where the second derivative of the filtered-based OMW reaches its maximum value, corresponding to the points of highest curvature between the peaks.

4) Detrended CP Signal Construction: The cuff deflation curve was interpolated at the identified foot points to create a “detrended CP signal,” representing a smooth, slowly varying pressure baseline free from superimposed oscillatory fluctuations.

5) Final OMW Generation: The pure oscillatory content was isolated by subtracting the detrended CP signal from the original cuff deflation curve, resulting in the “FootPoint-based OMW” [21]. This waveform retains the oscillatory features essential for PTT analysis while removing slow pressure trends. Notably, no additional filtering was applied, preserving the waveform’s temporal characteristics for accurate analysis. For further details on this method, readers are referred to [21].

C. Oscillometric Pulse Wave Decomposition

To extract the arterial FW and RWs, three Gaussian curves were fit to each oscillometric pulse, defined as the segment between two consecutive foot points identified during the “Foot Point Determination” step described earlier. Each Gaussian was characterized by three parameters: center, height, and standard deviation (SD). To ensure accurate fitting, we initialized these parameters using seven critical points on each oscillometric pulse. These points were identified with first and second derivatives and include the oscillometric pulse peak, maximum slope point, anacrotic notch, early systolic peak (FW peak), late systolic peak (first RW peak), dicrotic notch, and second RW peak (see Fig. 2).

The method for detecting these critical points closely resembles techniques used for photoplethysmography (PPG) critical points detection [27] was as follows.

1) *Oscillometric Pulse Peak:* Point with the largest amplitude.
2) *Maximum Slope Point:* Point of maximum amplitude in the first derivative.
3) *Anacrotic Notch:* Local second derivative maximum, located between the maximum slope point and 1/6th of the pulselength.
4) *Early SystolicPeak (FW Peak):* Point where the first derivative equals zero, the second derivative is negative, and post the maximum slope point and pre-anacrotic notch. In the absence of such a point, it is the local minimum of the second derivative after the maximum ascending slope and before the anacrotic notch.
5) *Late Systolic Peak (RW_1 Peak):* The first point posts the anacrotic notch where the first derivative is zero and the second derivative is negative. Alternatively, it is the first local minimum in the second derivative after the anacrotic notch.
6) *Dicrotic Notch:* Local second derivative maximum, located between the RW_1 and 1/8th of the pulselength.
7) *RW_2 Peak:* First point after the dicrotic notch where the first derivative is zero and the second derivative is negative.

These extracted characteristic points were used to initialize the parameters of three Gaussian curves fit to each oscillometric pulse for the extraction of the FW and two RWs (see Table I).

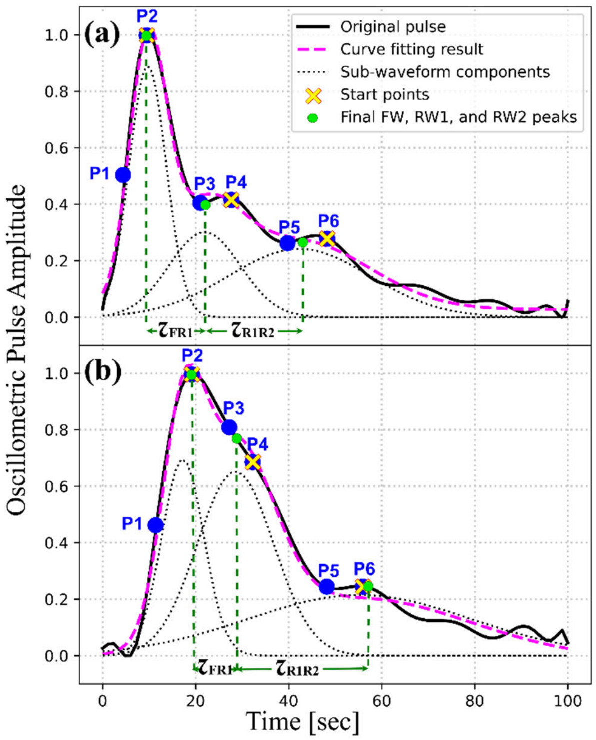

Fig. 2. GCF of oscillometric pulses. (a) Dataset 1: 46-year-old male. (b) Dataset 2: 65-year-old female. Original waveform (black), fit superposition (magenta dashed), and individual Gaussians (dotted black) are shown. Yellow crosses: start points. Green dots: final optimized values (LSE). Key points include maximum slope point (P1), oscillometric pulse peak (P2, FW Peak), anacrotic notch (P3), first RW peak (P4), dicrotic notch (P5), and second RW peak (P6).

TABLE I
INITIALIZATION OF THE GAUSSIAN PARAMETERS USING OSCILLOMETRIC CRITICAL POINTS

PARAMETER	INITIAL VALUE
AMPLITUDE OF THE FIRST GAUSS	Pulse amplitude at the oscillometric pulse peak location
AMPLITUDE OF THE SECOND GAUSS	Pulse amplitude at the first RW peak location
AMPLITUDE OF THE THIRD GAUSS	Pulse amplitude at the second RW peak location
CENTER OF THE FIRST GAUSS	The FW peak location
CENTER OF THE SECOND GAUSS	The first RW peak location
CENTER OF THE THIRD GAUSS	The second RW peak location
SD OF THE FIRST GAUSS	The interval between the FW peak and anacrotic notch
SD OF THE SECOND GAUSS	The interval between the first RW peak and dicrotic notch
SD OF THE THIRD GAUSS	The interval between the second RW peak and dicrotic notch

To improve BP estimation accuracy, we introduce a pulse wave decomposition approach using GCF. Direct identification of early and late systolic peaks in OMWs is unreliable due to noise, irregularities, and individual variability. GCF addresses these challenges by smoothing noise, adapting to waveform variability, and minimizing LSE. This method reduces sensitivity to fluctuations and avoids reliance on precise derivative calculations, ensuring accurate extraction of forward and reflected wave critical points. In our approach, each oscillometric pulse denoted by $\text{pulse}_{\text{omw}}$ is modeled as the superposition of three Gaussian curves representing the FW, first RW, and second RW as follows:

$$\text{pulse}_{\text{OMW}}(n,\theta) = \sum_{k=1}^{3} f_k(n,\theta) + O \quad n = 1,\ldots,N \tag{1}$$

with

$$f_k(n,\theta) = \frac{A_k}{W_k.\sqrt{2\pi}} e^{-\frac{(n-C_k)^2}{2W_k^2}} \quad n = 1,\ldots,N \tag{2}$$

where f_k is the kth Gaussian, defined by the parameter set θ that includes A_k, W_k, and C_k, representing the amplitude, SD, and center of Gaussian, respectively. Parameter O is an offset that aids in accurately modeling the OMW pulse. Here, n represents the position within the pulse wave, ranging from 1 to N, the length of each pulse. For $k = 1$, $k = 2$, and $k = 3$, f_k represents the FW, first RW, and second RW, respectively.

To fit these curves onto the pressure pulse, optimization of ten parameters, including A, W, and C for each of the three Gaussian curves, along with the offset, is necessary. Initial values for these parameters are based on critical points as outlined in Table I. The curve fitting problem is formulated as the minimization of the LSE, given by

$$\min_{\theta} \text{LSE} = \sum_{n=1}^{N} \left(\text{pulse}_{\text{org}}(n) - \text{pulse}_{\text{comp}}(n,\theta)\right)^2 \tag{3}$$

where $\text{pulse}_{\text{org}}$ denotes the observed pulse wave. The objective is to minimize the squared differences between observed data points and the values predicted by the GCF model, parameterized by θ. The optimization was performed using the Levenberg–Marquardt algorithm, which is well-suited for nonlinear least-squares problems.

Fig. 2 visually presents this pulse pressure decomposition method. The original pulse and the superposition of three fit Gaussian curves are shown in solid black and magenta dashed lines, respectively. The fit Gaussian curves, from left to right, represent the FW, the first RW, and the second RW. Their initial center values are marked in yellow crosses, while the final values determined using LSE are indicated in green dots. Notably, the peak locations identified through LSE closely align with those detected via derivative-based methods from the previous step. For a more robust approach, however, the peaks determined through pulse pressure decomposition were utilized in subsequent stages. Our pulse waveform decomposition method yielded a mean absolute percentage error of 0.18% and 0.16% across the first and second datasets, respectively.

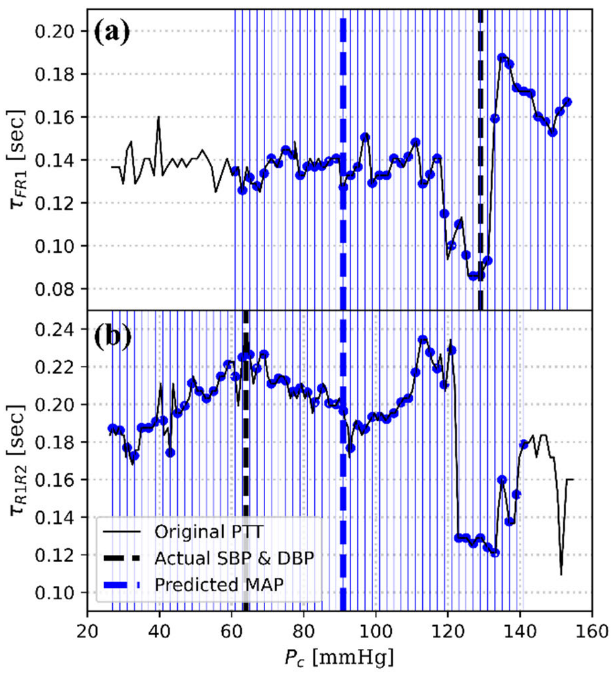

Fig. 3. Extracting features from PTT. (a) For SBP estimation, $\mathcal{Z}_{FR1}$ in range of $\text{MAP}_{\text{MAA}} - 30$ mmHg to $\text{MAP}_{\text{MAA}} + 110$ mmHg is evenly sampled at 2-mmHg increments of the CP. The actual SBP is displayed in a black dashed line. (b) For DBP estimation, $\mathcal{Z}_{R1R2}$ in range of $\text{MAP}_{\text{MAA}} - 90$ mmHg to $\text{MAP}_{\text{MAA}} + 50$ mmHg is evenly sampled at 2-mmHg increments of the CP. The actual DBP is displayed in a black dashed line. The MAP estimated using the MAA is displayed with a blue vertical dashed line.

D. PTT as a Feature in ML

Using the extracted FW and RWs, we defined two PTT patterns. The time interval between the peaks of the FW and the first RW (C_2 and C_1) is represented by $\mathcal{Z}_{FR_1}$, while the time interval between the peaks of the first and second RWs (C_3 and C_2) is denoted by $\mathcal{Z}_{R_1R_2}$. By mapping these PTTs against CP values, as illustrated in Fig. 3, two key observations were made: 1) $\mathcal{Z}_{FR_1}$ reaches a local minimum approximately at P_c = SBP and 2) $\mathcal{Z}_{R_1R_2}$ attains a local maximum approximately at P_c = DBP, where P_c represents CP as a function of time. While these mappings provide a direct pathway to estimate SBP and DBP, the intricate relationship between PTT and BP is influenced by factors such as pulse pressure, arterial capacity, and noise in the measured data. The measured PTT waveforms are often noisy, and their peaks do not consistently align with the SBP and DBP values. To address this complexity and the nonlinear nature of the data, ML algorithms were incorporated into the analysis.

The PTT samples, as a function of CP, were used as input features for ML algorithms. These algorithms were designed to model and estimate the relationship between BP and PTT. To ensure precise alignment of PTT measurements with corresponding CP values, a sampling strategy was implemented as follows:

$$\mathcal{Z}^i_{\text{SBP}} = \mathcal{Z}_{FR1} \text{ sampled at } (\text{MAP}_{\text{MAA}} - 30 + 2i) \tag{4}$$
$$\mathcal{Z}^i_{\text{DBP}} = \mathcal{Z}_{R1R2} \text{ sampled at } (\text{MAP}_{\text{MAA}} - 90 + 2i) \tag{5}$$

where $i = 0, \ldots, 70$ and MAP_{MAA} is the MAP determined using the maximum amplitude algorithm (MAA) [2]. MAA estimates MAP as the CP at which the maximum amplitude of the OMWE occurs. In our study, MAP serves as the central reference point for defining the CP ranges, representing an intermediate pressure between SBP and DBP. The ranges for SBP and DBP were based on their physiological relationships with MAP; in healthy individuals, SBP is typically 20–30 mmHg higher than MAP, and DBP is typically 10–20 mmHg lower. However, to ensure comprehensive coverage, account for intrinsic MAP estimation errors, and improve robustness across diverse conditions and populations, we significantly extended these ranges. Specifically, SBP is set to [$\text{MAP}_{\text{MAA}} - 30$, $\text{MAP}_{\text{MAA}} + 110$], and DBP is set to [$\text{MAP}_{\text{MAA}} - 90$, $\text{MAP}_{\text{MAA}} + 50$]. These expanded ranges were also visually verified in both datasets to confirm that the sampled segments consistently captured the relevant oscillometric pulse features necessary for reliable SBP and DBP estimation. Fig. 3 illustrates this procedure on a sample PTT waveform. In cases where CP values fell outside our selected range, a not a number (NaN) was used. The set generated from this approach, termed the "PTT Samples Dataset," is a matrix where each row corresponds to an individual subject and each column represents a feature. The features include 70 PTT samples and MAP_{MAA}, totaling 71 columns. This dataset forms the input to our ML models.

E. Missing Data Management

Not all ML algorithms inherently support missing values (NaNs). Therefore, we adopted a repetitive replacement approach, utilizing Bayesian ridge regression for imputation. NaNs were initialized using averaging, and features were imputed in ascending order of their missing value count. Initially, the "PTT Samples Dataset" is segmented into two subsets: X, the feature matrix, and y, the target vector, representing the feature column with missing values. Then, X is divided into two subsets: X_{known}, containing samples with known values, and X_{missing}, containing those with missing values for the target feature column, y. By training the regression model on X_{known} and the corresponding y_{known}, the missing values (y_{missing}) are predicted. Subsequently, these predicted values replace the missing entries in the original dataset. This imputation process is iteratively applied to each feature until it reaches either a maximum of ten iterations or meets a stopping condition based on the change in predictor values between consecutive iterations. A stopping criterion compares the maximum change in X values from one iteration to the next, defined as follows:

$$\frac{\max(|X_t - X_{t-1}|)}{\max(|X_{\text{known}}|)} < \text{tol}. \tag{6}$$

If this ratio falls below a predefined tolerance threshold (here, set to 10^{-3}), the process is halted.

It is important to mention that the NaNs in the test set were imputed using the regressor model fit on the training set, maintaining the integrity and separateness of the training and testing phases.

F. Data Normalization

Feature normalization is a crucial preprocessing step implemented to scale the data uniformly, essential for facilitating analysis and minimizing the impact of outliers. In this study, we specifically employed Z-score normalization. Z-score normalization standardizes every data point in a dataset so that the overall mean becomes 0, and the SD becomes 1. Each feature is individually normalized using its own mean and SD. This is achieved using the following formula:

$$\mathcal{Z}'^{(i,j)}_{\text{SBP}} = \frac{\mathcal{Z}^{(i,j)}_{\text{SBP}} - \mu_{\mathcal{Z}^{i}_{\text{SBP}}}}{\sigma_{\mathcal{Z}^{i}_{\text{SBP}}}} \tag{7}$$

where $\mathcal{Z}'^{(i,j)}_{\text{SBP}}$ and $\mathcal{Z}^{(i,j)}_{\text{SBP}}$ represent the standardized and original value, respectively, of the ith feature of the jth subject in the "PTT Samples Dataset," while $\mu_{\mathcal{Z}^{i}_{\text{SBP}}}$ and $\sigma_{\mathcal{Z}^{i}_{\text{SBP}}}$ indicate the mean and SD of the ith feature. It is crucial to note that during the normalization process for the test set, the mean SD of each feature is computed from the corresponding feature in the training set.

G. Machine Learning

In this study, we implemented four different ML algorithms to model the relationship between PTT and BP. These algorithms include the following.

1) *Random Forest:* A powerful ensemble tree-based ML algorithm that utilizes a bagging strategy to reduce variance.
2) *AdaBoost:* An ensemble tree-based algorithm that employs a boosting strategy to minimize bias.
3) *SVR:* A regression technique that utilizes kernel functions to model complex data patterns by minimizing margin violations within a defined tolerance margin.
4) *MLP:* A nonlinear technique employing multiple layers of nodes with weights and nonlinear activation functions.

To fine-tune model parameters and ensure unbiased error estimation, we employed a leave-one-out nested cross-validation method. This strategy involves two distinct loops.

1) Outer Loop: All data from one subject were reserved as the test set, while the remaining subjects' data formed the training set. This ensures that no data from the test subject were included in the training process, eliminating any overlap and minimizing overfitting.

2) Inner Loop: Fivefold cross validation was performed on the training data to optimize hyperparameters. The optimal parameters obtained from the inner loop were then applied to train the model on the training data, which was subsequently evaluated on the outer loop test set.

This method, though computationally intensive, is well-suited for smaller datasets and rigorously assesses the model's generalizability.

For the first dataset (ten individuals and 15 recordings each), the outer loop consisted of ten iterations, with each iteration reserving all 15 recordings from one individual as the test set and the remaining 135 recordings as the training set. For the second dataset (85 individuals and five recordings each), the outer loop included 85 iterations, with five recordings from one individual serving as the test set and the remaining 420 as the training set.

TABLE II
OPTIMAL PARAMETERS AND SEARCH REGIONS FOR DIFFERENT ML MODELS

Model	Parameter	Optimal Value	Search Region
Random Forest	#Estimators	80	50-100
	Criterion	MSE	MSE, MAE, or Poisson
	Max depth	Unrestricted Growth	Unrestricted Growth
	Min samples split	2	1-5
AdaBoost	#Estimators	80	50-100
	Base Estimator	Decision Tree	Decision Tree
	Learning rate	1.0	0.1-2.0
	Loss Function	Linear	Linear, square, exponential
MLP	Activation function	Relu	Relu, tanh, logistic
	Learning rate	0.001	0.0001-0.01
	Solver	Adam	SGD, Adam, or LBFGS
	Hidden Layer Size	100	50-200
SVR	Kernel	Polynomial	Polynomial or RBF
	C	0.1	0.01-10
	Gamma	0.01	0.01-0.5
	Epsilon	0.1	0.01-2

The inner loop used fivefold cross validation for hyper-parameter tuning, involving an initial random search of 50 iterations across a broad parameter range. Based on the random search results, a narrower search region was explored using a grid search to identify the optimal hyperparameters. The models trained with these parameters were then evaluated on the test set reserved in the outer loop. A detailed description of the optimal parameters and search regions for each ML method is provided in Table II.

III. EXPERIMENTAL RESULTS

Table III presents the outcomes of implementing the proposed techniques on two distinct oscillometric datasets: one comprising 150 upper arm recordings from ten individuals and the other consisting of 425 wrist recordings from 85 individuals. To ensure equitable evaluation, our approach was compared with methodologies applied to the respective datasets. For the initial dataset, reference SBP and DBP were acquired using the FDA-approved Omron HEM-790IT device, and for the second dataset, reference SBP and DBP were determined by averaging measurements taken by two nurses (see Section II-A for more details).

The performance was evaluated in terms of mean error (ME), mean absolute error (MAE), and SD of the error (SDE). It is noteworthy that SDE is deemed the most critical measure for comparison, as ME represents measurement bias that can be minimized with appropriate methods. For comparison, our methods were evaluated against established algorithms in the field: OMWE-CP [28], PTT-CP-ECG [22], and PTT-CP [21] for the first dataset and DBN-DNN [29], and feed-forward neural network (FFNN) [8] and PTT-CP [21] for the second dataset. The OMWE-CP method uses the OMWE for BP

TABLE III
COMPARISON OF THE PROPOSED BP ESTIMATION METHOD WITH PREVIOUSLY PUBLISHED WORKS ON DATASETS 1 AND 2

		Dataset 1							Dataset 2						
		OMWE-CP [28]	PTT-CP-ECG [22]	PTT-CP [21]	**PTT-GCF-RF**	PTT-GCF-AB	PTT-GCF-SVR	PTT-GCF-MLP	DBN-DNN [29]	FFNN [8]	PTT-CP [21]	**PTT-GCF-RF**	PTT-GCF-AB	PTT-GCF-SVR	PTT-GCF-MLP
SBP	ME	-1.23	3.09	**0.01**	-0.08	-0.30	-0.08	-0.25	**0.01**	-	**-0.01**	0.09	-0.80	0.47	0.12
	SDE	6.31	5.81	7.03	**3.71**	4.45	10.66	19.44	6.35	8.58	7.98	**6.42**	6.71	11.66	6.72
	MAE	4.95	5.31	6.22	**2.60**	3.42	8.48	11.88	-	6.28	6.39	**4.97**	5.28	9.11	5.28
DBP	ME	2.60	2.39	**0.01**	0.06	**-0.01**	-0.32	0.24	0.03	-	**0.01**	0.47	0.23	-0.40	-0.06
	SDE	7.38	5.78	5.96	**2.19**	2.55	7.39	13.79	5.28	7.33	5.39	**3.62**	3.73	8.84	4.33
	MAE	5.40	4.51	4.74	**1.59**	2.05	5.94	11.48	-	5.73	4.30	**2.80**	2.87	6.87	3.35

TABLE IV
BHS GRADING OF THE PROPOSED BP ESTIMATION METHODS

		Dataset 1				Dataset 2			
		PTT-GCF-RF	PTT-GCF-AB	PTT-GCF-SVR	PTT-GCF-MLP	**PTT-GCF-RF**	PTT-GCF-AB	PTT-GCF-SVR	PTT-GCF-MLP
SBP	Grade	**A**	A	D	D	**A**	B	D	B
	<=5	**84.6**	81.3	34.6	45.3	**60.7**	55	33.3	56.9
	<=10	**98.0**	96.0	60.6	66.6	**87.4**	85.3	62.3	85.8
	<=15	**100**	99.33	81.3	75.3	**97.1**	96.6	80.3	96.2
DBP	Grade	**A**	A	C	D	**A**	A	C	A
	<=5	**97.3**	96.0	47.3	26.6	**83.6**	83.4	45.4	78.2
	<=10	**100**	**100**	81.3	46.6	**98.5**	98.5	75.1	96.2
	<=15	**100**	**100**	96	65.3	**100**	**100**	90.2	99.7

estimation. The PTT-CP-ECG method estimates BP based on the pattern of PTT obtained from simultaneous oscillometric and ECG recordings. The PTT-CP approach estimates BP from PTT using only the oscillometric recordings. Our approach, PTT-GCF-ML, estimates BP from PTT extracted directly from oscillometric recordings using GCF and ML (RF, AB, SVR, and MLP). It is worth noting that the results of these methods were obtained using the same dataset as ours, ensuring fair comparison. To further analyze the performance of the proposed method compared to the reference measurements, the Bland–Altman analysis was also performed.

Our proposed PTT-based method leveraging random forest, AdaBoost, and MLP as ML regressors demonstrates superior performance. As shown in Table III, our methods achieve notably low SDEs and minimal biases. Among the ML methods, random forest stands out. For the first dataset, it achieved an SDE of 3.71 mmHg for SBP and 2.19 mmHg for DBP. For the second dataset, the SDE was 6.42 mmHg for SBP and 3.62 mmHg for DBP, highlighting its robustness in providing precise BP estimations using the PTT pattern.

Compared to the baseline PTT-CP method, integrating random forest significantly reduced the SDE. For the first dataset, SDE for SBP decreased by 47% and for DBP by 63%. In the second dataset, the SDE improved by 19% for SBP and 33% for DBP. These results demonstrate the enhancements in BP estimation accuracy achieved by combining ML with the PTT-CP method.

While the PTT-CP method has lower ME due to a bias correction step, our method achieves significantly lower SDE, capturing overall variability and precision more effectively. ME, reflecting systematic bias, can be addressed through calibration or post hoc bias correction techniques. Future work will explore incorporating such corrections to further improve ME while maintaining the high precision of the reduced SDE.

It is noteworthy that the ME and SDE for the proposed methods, utilizing random forest, AdaBoost, and MLP, were consistently less than 5 and 8 mmHg, respectively. This achievement demonstrates compliance with the AAMI standard [23] for both SBP and DBP estimation. Additionally, our method utilizing random forest achieved a Grade A ranking according to the BHS criteria [24] (see Table IV). Consequently, our proposed methods hold significant potential for integration into the next generation of oscillometric BP devices, leading to more reliable and accurate BP measurements.

The Bland–Altman plots in Figs. 4 and 5 illustrate a narrow range of agreement between estimated and actual SBP and DBP measurements. Notably, the greatest scatter for both SBP and DBP occurs at the higher and lower ends of the pressure range, with tighter agreement observed at mid-range pressures. This pattern may be attributed to the limited number of participants with high DBP in the dataset. Among the ML algorithms, SVR exhibited the most pronounced overestimation of low BP values and underestimation of high BP values, followed by MLP. In contrast, AdaBoost and random forest performed more favorably across the entire BP range, with random forest achieving the most consistent results.

IV. DISCUSSION

Monitoring BP is crucial for cardiovascular health assessment and disease management. In this study, we presented a novel method to improve BP assessment accuracy using a multistep approach: extracting the OMW, decomposing oscillometric pulses into FW and RWs through derivative

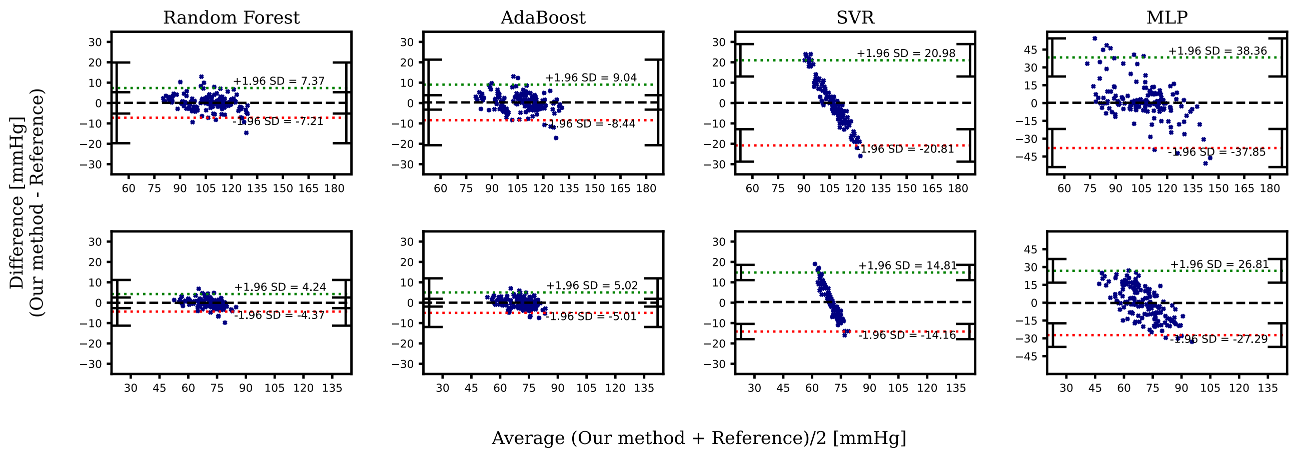

Fig. 4. Bland–Altman plots of the SBP and DBP estimates for Dataset 1 using different ML algorithms.

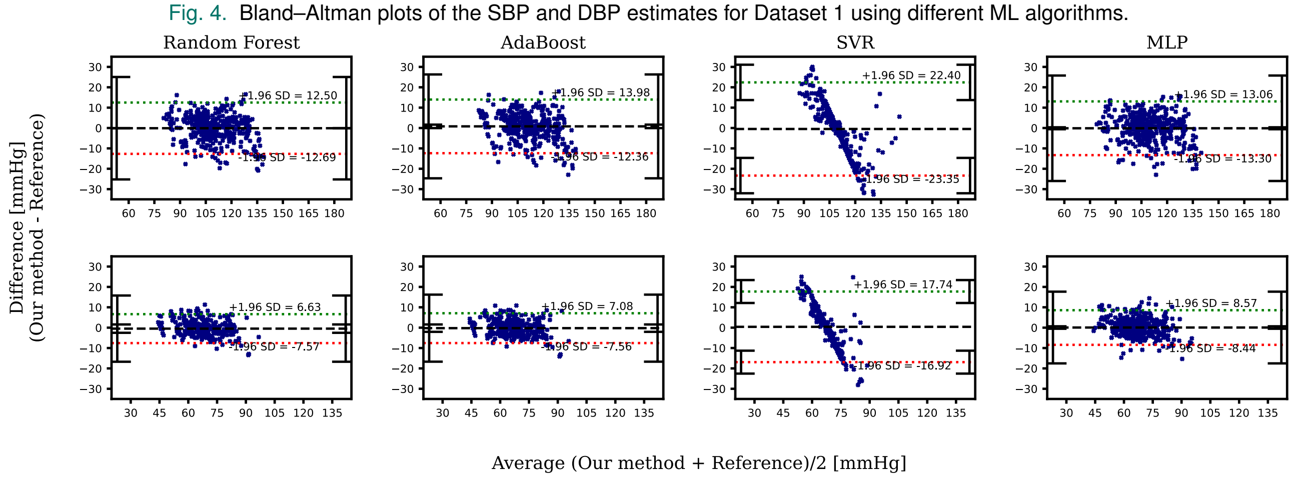

Fig. 5. Bland–Altman plots of the SBP and DBP estimates for Dataset 2 using different ML algorithms.

analysis and GCF, and refining our approach with ML based on PTT-CP mappings. Validation across two distinct datasets showed that when employing ML algorithms such as random forest, AdaBoost, and MLP, the proposed method met the AAMI standard for both SBP and DBP estimation. This achievement underscores the reliability and accuracy of our methods and suggests their potential integration into new oscillometric BP devices for enhanced BP estimation.

Among the ML algorithms tested, random forest and AdaBoost consistently demonstrated superior performance, likely due to their ability to handle nonlinear relationships and robustness to outliers—common challenges in oscillometric BP estimation. The MLP method, however, exhibited varying performance across datasets, underperforming on the first but achieving satisfactory results on the second. This discrepancy may be attributed to differences in dataset sizes and the specific architecture and hyperparameter tuning of the MLP model. Neural networks like MLP require large, diverse training datasets and careful optimization for optimal performance. Enhancing the MLP model by increasing the number of records and refining its architecture could potentially improve its results significantly.

The observed performance variability between datasets highlights the need for a more extensive and diverse training dataset to achieve consistent results across different patient populations and physiological conditions. Additionally, the datasets available for cuff-based oscillometric BP estimation are relatively limited in size, posing challenges in fully generalizing our findings to broader populations. Expanding the scope to include larger and more diverse datasets specific to cuff-based methods will be crucial for future research and will significantly enhance the robustness of our conclusions. To address these limitations, our study employed a nested cross-validation strategy, providing a robust framework for assessing generalization within the available datasets. While this approach effectively evaluates the model's performance, future work could explore validation on independent external datasets to further establish the model's reliability and applicability across broader populations.

Our method involves computational demands due to GCF, but ECG-based PTT techniques also require substantial processing for tasks like R-peak detection and multisignal synchronization. By relying on a single pressure signal, our approach simplifies the computational pipeline, eliminates multisensor alignment errors, and reduces hardware requirements, production costs, and potential timing discrepancies. Furthermore, modern wearable processors efficiently handle these computations, making the tradeoff between computational complexity and hardware simplicity advantageous.

An ablation study demonstrated that applying GCF generally improves BP estimation accuracy. The study compared SDE in mmHg for SBP and DBP estimation with and without GCF. For instance, with the random forest model in Dataset 1, SDE for SBP decreased from 4.06 to 3.71 mmHg, and for DBP, it decreased from 3.06 to 2.19 mmHg. Similar improvements were observed across other models and in Dataset 2, indicating that GCF enhances the robustness of PTT extraction, particularly in challenging conditions with noisy or irregular waveforms.

While our current study focused on pulse timing information and ML algorithms, future research could explore the incorporation of additional features, such as patient demographics (e.g., age, sex, and BMI) and other physiological data (e.g., heart rate and arterial stiffness), to potentially enhance BP estimation accuracy. Moreover, integrating our approach with existing PTT techniques using unobtrusive sensors [30] offers a promising avenue for achieving greater robustness and versatility. By fusing complementary methods, we can leverage the strengths of each approach to address challenges such as motion artifacts, signal synchronization, and variability across populations.

V. Conclusion

This study presented a novel approach for oscillometric BP estimation that leveraged pulse timing information enhanced with ML. By focusing on PTT patterns derived from decomposed pulse waveforms, our method overcame the limitations of traditional amplitude-based approaches. The promising results, meeting AAMI and BHS standards and demonstrating improved accuracy in two distinct datasets, highlighted the potential of this technique for enhancing the reliability and accuracy of oscillometric BP devices. Future work will focus on validating the method with larger, more diverse datasets and refining the peak detection algorithms to ensure robustness across various physiological conditions.

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